

Gaydar: Estimating Local LGBTQ Populations

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Abstract

How many LGBTQ people live in your neighborhood? Measurement of the LGBTQ community is notoriously challenging, particularly with any degree of geographic granularity. I produce tract-level estimates of non-binary (NB) and lesbian and gay, bisexual, transgender, and queer (LGBTQ) populations from publicly available US Census Bureau data. Foremost, I capture the spatial variation in the levels and concentration of NB and LGBTQ people, with separate estimates by gender identity for lesbians and gay, bisexual, and queer people. Hence, the estimates incorporate within-cohort variation in sexual orientation across different gender identities. I also capture local variation in the cross-sectional composition of the LGBTQ community (for example, bisexuals and queers comprise more of the community in college-student heavy tracts). Finally, I incorporate measures of uncertainty into the estimates, reflecting sampling variation across multiple Census data products. The estimates offer insights for scholars of LGBTQ demographics, public health practitioners, and myriad others.

Gaydar explores the spatial distribution of non-binary (NB) and/or gay and lesbian, bisexual, transgender, and queer (LGBTQ) people. I focus on adult (18+) NB and LGBTQ (sorry asexuals), split into three mutually exclusive and collectively exhaustive gender identities: men, women, and non-binary people.

To view the results of the estimation methodology described below, use this [interactive map](#).

Note transgender people might also identify as gay, lesbian, bisexual, or queer, meaning the sub-population shares will not sum to the total because I don't double count LGBTQ transgender people. Additionally, limitations within Household Pulse Survey microdata mean that I cannot measure non-binary transgender individuals.

Summary

- Estimate LGBTQ identification rates by state $\times$ sex assigned at birth $\times$ age bin using pooled Household Pulse Survey (HPS) microdata, with survey-weighted estimation and soft calibration to recent Gallup national benchmarks.
- Estimate gender identity composition conditional on sexual orientation, sex assigned at birth, and age, using HPS microdata to recover age-specific distributions of different gender identities. I assume cisgender, non-binary, and transgender are mutually exclusive and collectively exhaustive. Step 2 constructs a crosswalk from ACS age-sex counts to age-gender identity-sexuality estimates. These composition estimates are stabilized using age-local Dirichlet smoothing, allowing gender identity shares to vary smoothly across cohorts while remaining responsive to well-identified age groups (For example, younger AFAB lesbians roughly 15 percentage points more likely to identify

as non-binary than AMAB gays in the same age cohort. Age-local Dirichlet smoothing preserves the pattern, just smoothing out some of the noise).

- Estimate LGBTQ sub-population shares (i.e. lesbians and gays, bisexuals, queers, transgender people) by pooling nationally rather than by state. State-level cells (think transgender people in any state or bisexuals in Wyoming) were too sparse to trust even with empirical Bayes shrinkage across states, so every state now applies the same national LG/bisexual/queer mix to its own state-level identification rate from step 1.
- Project expected LGBTQ population counts to Census tracts by combining estimated identification rates with ACS 5-year age-sex population counts (table B01001). Since age and sex vary locally, sexuality and gender identity will too.
- Reweight projected LGBTQ populations within states by blending two PUMA-level spatial proxies: the observed concentration of same-sex couples, and the observed concentration of singles within the same birth sex and age cell. Note that same-sex couples tend to be older, wealthier, better educated, and more likely to identify as gay or lesbian than the broader LGBTQ population (Julian et al., 2024), which suggests their spatial concentration patterns might systematically differ — especially for young, non-binary, and transgender people, who are least likely to show up as a Census-visible couple at all (Gates, 2013). I address this by blending in the concentration of singles of the same sex and age, weighted by that cell’s HPS-estimated marriage rate: cells that are more often married lean on the couples signal, cells rarely married lean on the singles signal instead. Nonetheless, the reweighting ensures the estimates reflect more than the age and sex composition of a Census tract. The reweighting produces a calibration factor for each PUMA, but applying that factor as a hard step function at PUMA boundaries introduces visual discontinuities in the map that are artifacts of the PUMA delineation rather than real demographic transitions. To address this, I smooth the calibration factors to the tract level using inverse-distance weighting (IDW): each tract receives a distance-weighted blend of all PUMA factors rather than the factor of whichever PUMA it happens to fall in. This preserves the geographic heterogeneity the calibration is meant to capture while producing a spatially continuous surface.
- Enforce a capacity constraint so that no tract’s LGBTQ estimate exceeds its gender-group population. In high-concentration areas, the reweighting and rescaling can push estimates above 100 percent of the relevant gender-group total. I cap each gender group’s estimates at its population and scale sub-populations (lesbian/gay, bisexual, queer, transgender) down proportionally, preserving within-group composition.

Data Sources

Household Pulse Survey (HPS)

HPS was the first the Census data product to ask respondents about sexual orientation and gender identity, basing the design of questions on the results of a 2022 NASEM report (National Academies of Sciences, Engineering, and Medicine, 2022). Designed to elicit responses to rapidly changing situations (i.e. food security early in the coronavirus pandemic), the HPS surveyed different households each week. While topics varied between weeks, HPS consistently collected the following demographic data:

- Sexual orientation
- Gender identity
- Sex assigned at birth
- Year of birth
- Marital status
- State of residence
- Survey weights

The microdata allow the relationship between sexual orientation and gender identity to vary flexibly by age and geography. For example, younger gay and lesbian individuals are more likely to identify as non-binary than older cohorts.

Gallup Polling

The February 2026 Gallup report provides the most recent nationally representative estimate of LGBTQ identification in the United States.

The report includes:

- Overall national LGBTQ identification
- LGBTQ identification by generation
- LGBTQ identification by generation $\times$ sex (for select cohorts)

Given uncertainty in survey measurement, these estimates are used as soft calibration targets, not hard constraints. In practice, most state-level estimates fall within one to two percentage points of Gallup’s top-line estimate of 9 percent.

American Community Survey (ACS)

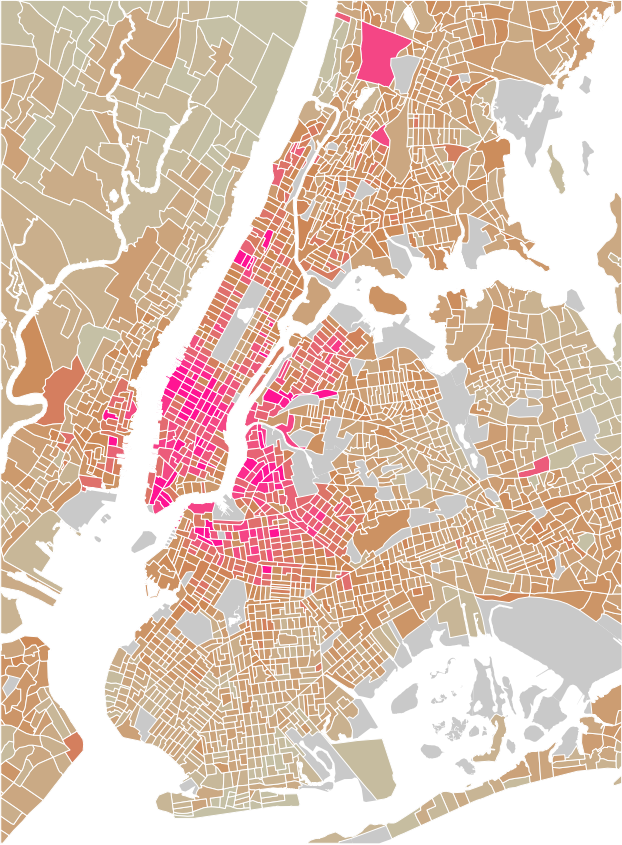
Population projections are anchored to ACS 5-year estimates, which provide stable small-area population counts at fine geographic resolution.

Specifically, the pipeline uses:

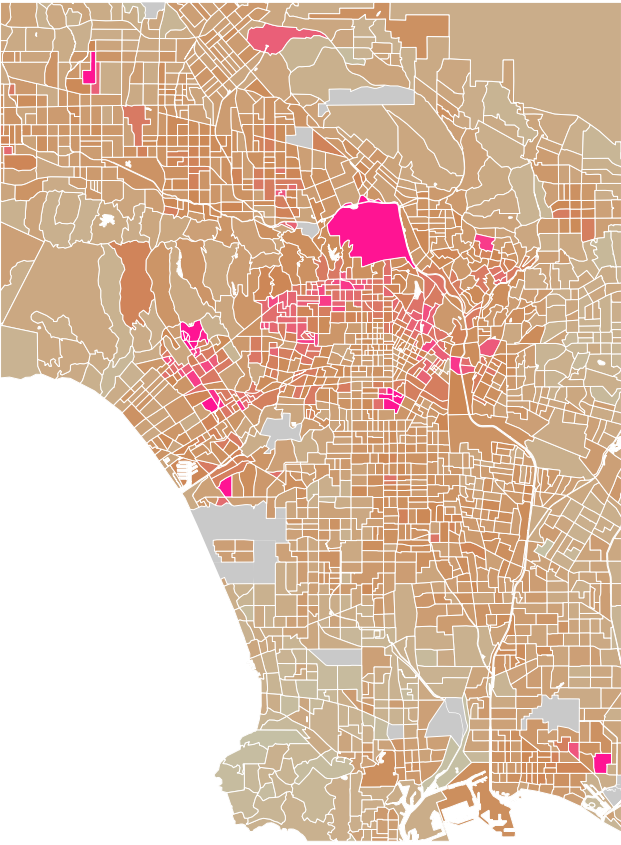
- Table B01001 (Sex by Age) to obtain population counts by sex assigned at birth and granular age bins, measured at the tract level.
- Table B09019 (Household Type by Relationship) informs the same-sex-couples half of the spatial reweighting above, measured at the PUMA level.
- Table B12002 (Sex by Marital Status by Age) provides the singles-concentration signal blended into the same reweighting, also measured at the PUMA level.

Because the Household Pulse Survey (HPS) and ACS use different age groupings, HPS respondents are mapped into ACS-compatible age bins prior to estimation. This ensures that estimated LGBTQ identification rates and gender-composition shares can be directly applied to ACS population counts without interpolation or extrapolation.

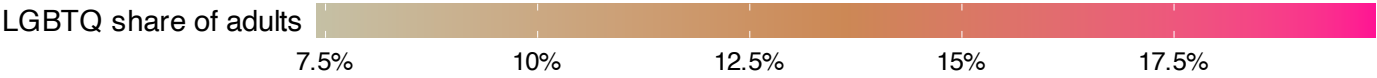
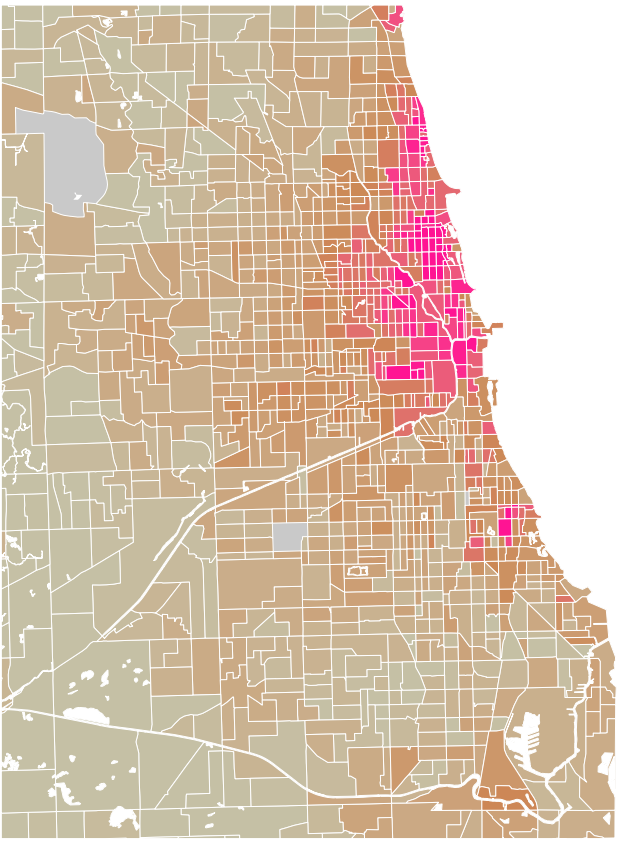
New York



Los Angeles



Chicago



References

- Gates, Gary J.**, “Geography of the LGBT Population,” in Amanda K. Baumle, ed., *International Handbook on the Demography of Sexuality*, Vol. 5 of *International Handbooks of Population*, Dordrecht: Springer, 2013, pp. 229–242.
- Julian, Christopher A., Wendy D. Manning, and Krista K. Westrick-Payne**, “Responses to Sexual and Gender Identity Measures in Population-Level Data by Birth Cohort: A Research Note,” *Demography*, 2024, *61* (1), 15–30.
- National Academies of Sciences, Engineering, and Medicine**, *Measuring Sex, Gender Identity, and Sexual Orientation*, Washington, DC: The National Academies Press, 2022.